From Rapids to Floodplains: Ecological Boundaries Drive Genomic Divergence Without Morphological Change in Electric Eels

Preliminary report of ddRAD analyses by David de Santana

Recent phylogenomic and ecological analyses have revealed unexpected complexity in the evolutionary history of electric eels, previously considered a monotypic genus for over two centuries. We integrated genome-wide SNP data (>110,000 loci, subsequently refined using various filtering criteria), mtDNA divergence, and ecological variables to test species delimitation and speciation mode hypotheses across over 300 individuals sampled throughout Greater Amazonia. Our results support the existence of nine evolutionarily distinct lineages, partitioned into two major ecotypes, upland (shield headwaters) and lowland (floodplain rivers), which correspond to distinct environmental regimes and evolutionary processes. Phylogenetic and population structure analyses (IQ-TREE, SNAPP, PCA, STRUCTURE) confirm deep divergence among upland populations ($F_{st} = 0.30–0.67$; mtDNA K2P = 2.4–7.6%), consistent with Miocene drainage fragmentation and long-term allopatry across Guiana and Brazilian shield systems. In contrast, lowland lineages exhibit shallower divergence but strong ecological structuring. PCA of SNPs revealed three discrete genomic clusters corresponding to *E. multivalvulus* (Lineage 7), *E. varii* (Lineage 8), and Lineage 9, an undescribed blackwater-restricted lineage from the Negro basin. STRUCTURE admixture analysis ($K = 3$) further supports these divisions. Redundancy analysis (RDA) revealed strong ecological differentiation among lowland taxa, with habitat-associated SNPs distinguishing blackwater, whitewater, and mixed-water lineages. RDA1 explained a significant proportion of constrained genomic variance ($F = 11.02$, $p = 0.001$), reinforcing ecological speciation across water-type gradients. Strikingly, all nine lineages maintain a conserved external morphology, reflecting the stabilizing selection imposed by their extreme electrogenic anatomy. Together, these findings uncover four modes of speciation in *Electrophorus*: 1, ecological between upland and lowland ecotypes; 2, allopatric in the upland lineages; 3. ecological with gene flow across whitewater–blackwater gradients, and 4. cryptic speciation between *E. multivalvulus* and *E. varii*, with limited introgression. All unfolding beneath the surface of morphological stasis. These preliminary results highlight how ecological divergence, rather than geography or phenotype alone, can drive and maintain species boundaries in one of the most iconic vertebrates.

Background. *Electrophorus,* which was previously thought to consist of only one species, has now been shown to include at least three distinct species based on several evidence (de Santana et al., 2019). Recently, DNA barcoding (unp. data) suggests the existence of multiple lineages (Table 1 and Figure 1). Occupying a vast range across the Amazon, Orinoco, and Guiana Shield drainages, eels are notable for their ability to generate electricity, with discharges reaching as high as 860 volts. Despite these physiological novelties, little was known about the evolutionary forces shaping their diversification. Our recent DNA barcoding analyses identified eight lineages of electric eels (six in the upland and two in the lowland). Here, we use a phylogenomic dataset and ecological information to test three primary hypotheses: (1) Do major mtDNA lineages correspond to independently evolving species? (2) Does ecological divergence between upland and lowland habitats drive speciation? (3) Does morphological stasis mask cryptic diversification?

Table 1. K2P genetic divergence among the eight mtDNA genomic lineages.

Lineages	L1	L6	L3	L5	L4	L2	L8	L7
E_electricus (L1)								
E_voltai (L6)	0.06599							
E_upper_Madeira (L3)	0.06822	0.02438						
E_upper_Tocantins (L5)	0.06949	0.03661	0.03524					
E_upper_Maroni (L4)	0.07630	0.05124	0.04042	0.04090				
E_Orinoco (L2)	0.06727	0.04626	0.03241	0.03968	0.04247			
E_varii (L8)	0.09202	0.09067	0.10296	0.09776	0.09280	0.10019		
E._multivalvulus (L7)	0.09959	0.09628	0.10386	0.10251	0.09703	0.10138	0.00734	

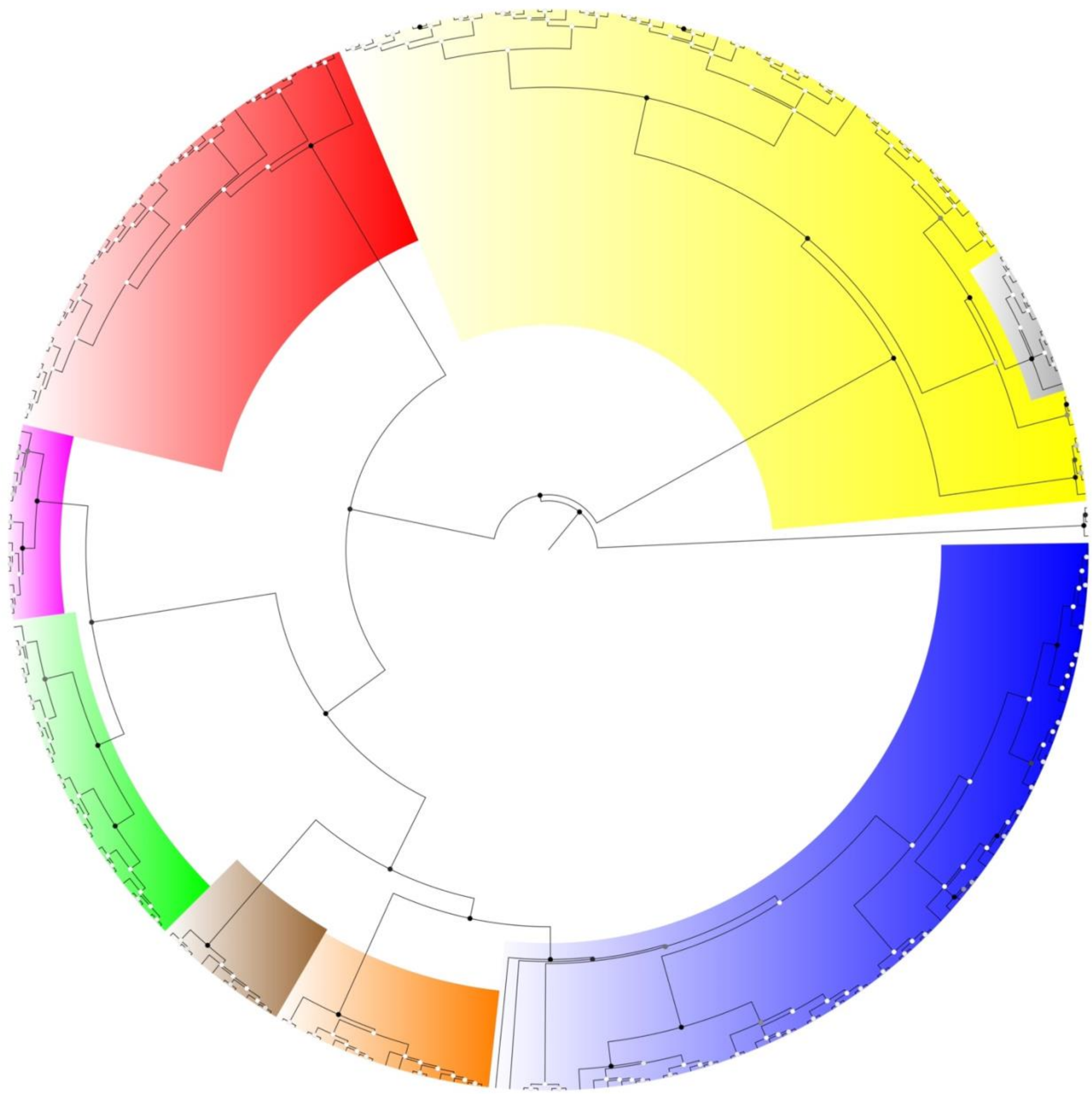


Figure 1. Bayesian Inference COI 701 bp and posterior probability (PP) for 302 samples of *Electrophorus* (eight lineages) and three outgroups. Color represent mtDNA lineages: red = *E. electricus* (Lineage 1): Guianas coastal rivers; clearwater; pink = Lineage 2: Orinoco, clear/blackwater; orange = Lineage 3: Upper Madeira, clearwater; Lineage 4: Upper Maroni; white/clearwater; brown = Lineage 5: Upper Tocantins; clearwater; blue = *E. voltai* (Lineage 6): Brazilian + Guiana Shields, mixed waters; balck = *E. multivalvulus* (Lineage 7), Ucayali/Pacaya, whitewater headwaters; *E. varii* (Lineage 8), whitewater floodplains. All lineages supported with a PP = 1.

Methods

***Sample Preparation and Sequencing*.** Marina Loeb sequenced samples at the Cornell Lab of Ornithology under the supervision of Casey Dillman. Genomic libraries were prepared using the restriction enzymes PstI (5'-CTGCAG-3′) and MspI (5'-CCGG-3′), following the protocol of Bangs et al. (2018).

***De Novo Assembly and SNP Calling*.** Raw reads were processed using the Stacks v2 pipeline in Galaxy and Python, including ustacks, cstacks, tsv2bam, gstacks, and populations modules**. A** de novo assembly was performed for the full dataset and separately for the lowland**,** upland**,** and combined datasets with and without outgroups**.** Parameters used in ustacks included**:** Minimum depth of coverage per stack: 3; Maximum distance allowed between stacks: 2 Maximum number of gaps allowed between stacks: 2

SNP Filtering and PCA. SNP datasets were filtered to remove low-quality loci and potential cross-contamination. Filters applied included: Minor Allele Frequency (MAF) threshold**:** 0.05 Maximum missing data per locus**:** 20% Hardy-Weinberg Equilibrium threshold**:** 1e-6 Minimum depth per SNP: 3.

The Principal Component Analysis (PCA) was conducted using custom R scripts (de Santana**)** to explore genomic structure. VCF files were converted into Phylip format in R to validate locus alignment, and results were also visualized using TESS3 v5 to generate UPGMA trees.

Phylogenetic Analyses

Filtered SNPs were exported to Phylip format using VCF2Phylip and Python scripts. Two types of phylogenetic analyses were conducted: **Maximum Likelihood (ML).** ML phylogenies were inferred using IQ-TREE, with the best-fit model selected by ModelFinder. Support was assessed with 1,000 ultrafast bootstraps, and consensus trees were visualized in FigTree v1.4.4. **Species Tree Inference (SNAPP)**.*Biallelic loci were extracted using custom R scripts (de Santana) and used as input for SNAPP in BEAST2. Given SNAPP's computational intensity, analyses were limited to ≤25 representative samples***.**

Population Structure and Admixture (STRUCTURE). To assess genetic admixture, VCF files were converted to STRUCTURE input format in R (see Supplementary Appendix XX), and analyses were run under a range of K-values to evaluate population clustering.

Redundancy Analysis (RDA). Ecological correlates of genomic variation were tested using Redundancy Analysis (RDA). Each sample was assigned to ecological categories including: Habitat (upland vs. lowland); Breeding season (rainy vs. dry); Sociality (social vs. solitary); Water type (blackwater vs. whitewater) RDA and SNP outlier detection were performed in RStudio using a suite of packages and newly developed code to assess ecological selection and habitat-associated genomic divergence.

R Packages

The following R packages were used for SNP filtering, PCA, population structure analysis, phylogenetic file conversion, and RDA:

- vcfR – VCF file import and manipulation (Knaus & Grünwald, 2017)
- adegenet – Multivariate analysis of genetic markers (Jombart, 2008)
- poppr – Population genetic analysis of clonal and sexual populations (Kamvar et al., 2014)
- pegas – Population and evolutionary genetics analysis (Paradis, 2010)
- SNPRelate – SNP data analysis and PCA from GDS format (Zheng et al., 2012)
- dplyr, data.table, stringr – Data manipulation and string processing
- ggplot2 – Data visualization
- vegan – (for RDA, not yet listed but likely used; add if applicable)

Bioinformatics and Phylogenetics Software

- **Stacks v2.65** – For de novo RADseq assembly and SNP calling (Catchen et al., 2013)
- **IQ-TREE v2.2+** – Maximum Likelihood phylogenetic inference with ModelFinder and ultrafast bootstrap (Nguyen et al., 2015)
- **VCF2Phylip** – VCF to Phylip file conversion for phylogenetic analyses
- **BEAST2 + SNAPP** – Species tree inference using biallelic markers (Bryant et al., 2012)

- **FigTree v1.4.4** – Visualization of phylogenetic trees
- **TESS3 v5** – Clustering and visualization of spatial genetic structure

Results

Upland and Lowland Clade Divergence. Phylogenomic analyses recovered a deep basal split between the upland and lowland *Electrophorus* lineage*s*, corresponding to distinct ecological regimes and speciation histories. Upland lineages, restricted to shield headwater drainages, showed high genomic divergence (Fst = 0.30–0.67) and deep mitochondrial divergence (mtDNA K2P = 2.4–7.6%), consistent with long-term allopatric divergence across ancient Miocene barriers. In contrast, lowland lineages exhibited shallower divergence, but were strongly differentiated by habitat and ecological traits. Ecological comparisons revealed marked contrasts between clades. Upland *Electrophorus* inhabit clearwater and mixed-shield rivers, breed during the rainy (high-water) season, and are social as adults, displaying coordinated hunting behaviors. They also show strong philopatry, returning annually to the same dry-season hunting sites, as observed in the Iriri River (e.g., Bastos et al., 2021, and observation after that). These traits likely reinforce reproductive isolation and promote population divergence within geographically structured upland systems, typically characterized by rapids. In contrast, lowland lineages such as *E. varii*, *E. multivalvulus*, and Lineage 9 occupy floodplain habitats, including whitewater (várzea) and blackwater (igapó) systems. These species breed during the dry season (lower water) and are generally solitary as adults, with juveniles forming groups only during early developmental stages. Behavioral differences coincide with contrasting hydrological regimes and are associated with ecological selection pressures in chemically distinct environments. This deep ecological partitioning between upland and lowland eels supports the hypothesis of ecological speciation, with major reproductive, behavioral, and habitat-related traits diverging along environmental gradients. The basal split observed in the phylogenies is thus not only genetic but strongly ecologically structured, providing a foundation for the independent diversification pathways observed in each clade.

The Redundancy analysis (RDA) revealed a strong genomic signal of ecological differentiation between upland clearwater/blackwater small rivers, streams, rapids, and lowland floodplains (várzea and Igapó) as a key driver of genetic differentiation in electric eels. The first RDA axis (RDA1), which accounted for 55.65% of the constrained genetic variance, clearly separated individuals by habitat type, underscoring the influence of environmental factors on genomic divergence (Figure 2). An ANOVA of the RDA model confirmed this effect was highly significant ($F = 170.66$, $p = 0.001$). Lowland populations formed a tight cluster, suggesting relatively homogeneous genomic backgrounds, likely reflecting recent divergence or ongoing connectivity across floodplain environments. In contrast, upland populations showed greater dispersion along both RDA1 and the first unconstrained axis (PC1, 11.16%), possibly due to deeper evolutionary histories or local adaptation within heterogeneous shield river systems. Analysis of SNP loadings on RDA1 identified 242 candidate loci in the top 5% of habitat-associated outliers, including 155 linked to lowland and 87 to upland environments (Figure 3). This asymmetry may reflect differences in the strength or uniformity of selective pressures across habitats. Floodplain environments are more geologically dynamic and seasonally extreme, with recurrent selective regimes (e.g., predation pressure), which may promote strong, parallel adaptation across populations, leading to more consistently differentiated loci. In contrast, upland habitats are more stable but heterogeneous, and selection may act in more localized or lineage-specific ways, resulting in fewer shared genomic outliers across upland populations. The top-ranked SNPs spanned both habitat types, with the strongest lowland-associated markers (e.g., SNPs 15996:137, 1600:98, 3321:97) showing RDA1 loadings exceeding 0.40, while upland-associated SNPs (e.g., 16385:35, 1848:74, 608:51:00) had loadings below –0.36. The presence of strongly differentiated SNPs in both directions supports a scenario of reciprocal adaptation, where distinct selective pressures in each habitat type shape genomic divergence. Altogether, these results provide compelling evidence that ecological factors, particularly habitat differences between floodplains and upland tributaries, are key drivers of genomic divergence and likely contribute to ongoing ecological speciation in this group. When aligned to the whole genome, we may be able to identify the names of the outliers, as speculated in Table 2.

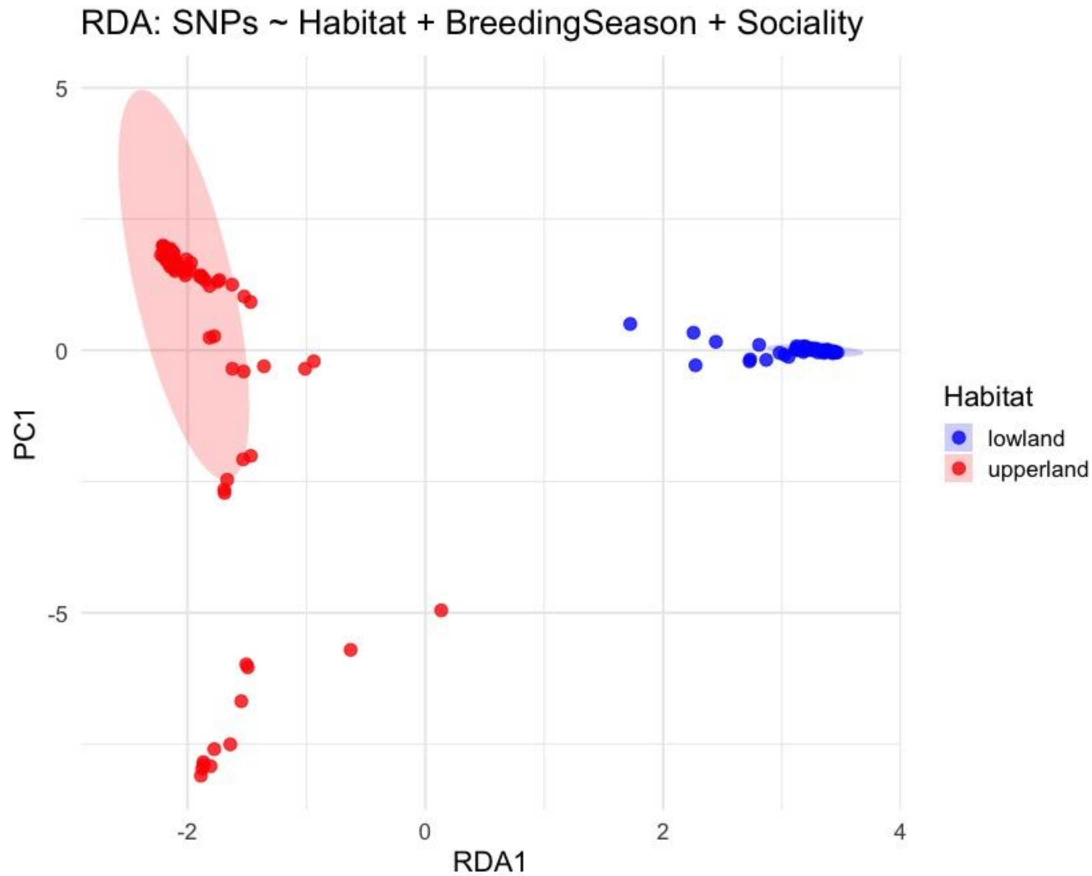


Figure 2. Redundancy analysis (RDA) of SNP variation constrained by habitat, breeding season, and sociality. Points represent individual samples, colored by habitat type: upland (red) and lowland (blue). The first RDA axis (RDA1) explains 100% of the constrained variance and separates individuals primarily by habitat. Upland individuals show greater dispersion along both RDA1 and the first unconstrained axis (PC1), indicating genomic heterogeneity, while lowland individuals cluster tightly, suggesting a more homogeneous genetic background. The shaded ellipse represents the 95% confidence interval for upland samples. This pattern supports a model of ecological divergence between floodplain and upland environments.

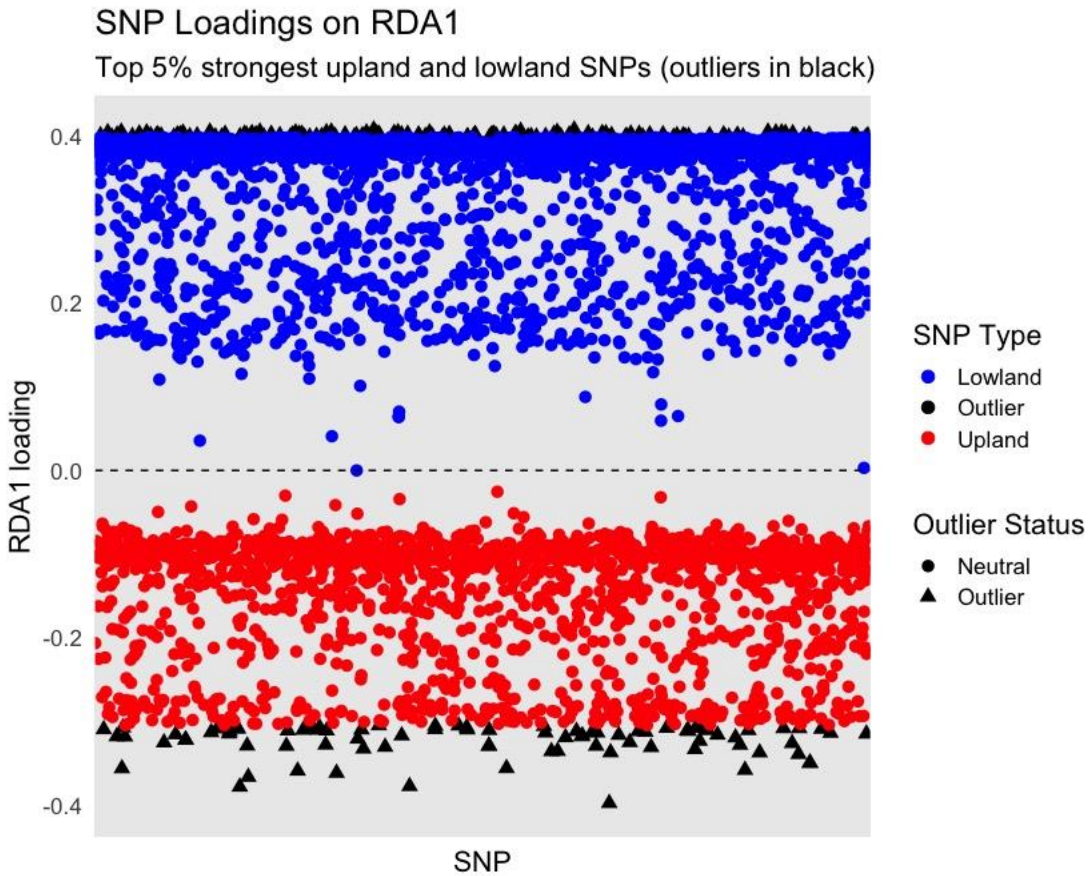


Figure 3. SNP loadings along the first constrained axis (RDA1) from the redundancy analysis (RDA), highlighting the top 5% most habitat-associated SNPs. Each point represents a single SNP, with color indicating habitat association (blue = lowland, red = upland), and shape indicating outlier status (black triangles = top 5% outliers). SNPs with positive RDA1 loadings are associated with lowland populations, while negative values indicate association with upland populations. The asymmetrical distribution of outliers (155 lowland, 87 upland) suggests stronger or more uniform selective pressures in floodplain environments. These loci are strong candidates for habitat-specific divergent selection and potential drivers of ecological speciation.

Table 2. Predicted functional categories and ecological relevance of candidate genes under divergent selection between lowland and upland *Electrophorus* lineages. This table summarizes hypothesized gene functions and ecological roles associated with habitat-specific adaptation in electric eels, based on RDA-identified SNP outliers and their putative alignment to functional categories. Candidate genes are grouped into ten categories linked to key physiological, behavioral, and ecological processes, including respiration, osmoregulation, reproduction, electric organ discharge modulation, and sensory processing. The third and fourth columns outline the hypothesized ecological roles of each gene category in lowland floodplain environments (e.g., blackwater and whitewater systems with hypoxia, high pathogen load, and seasonal inundation) versus upland shield rivers (e.g., fast-flowing clearwaters with high oxygen and structural complexity). These functional hypotheses provide a framework for future whole-genome annotation in *Electrophorus*.

Functional Category	Hypothetical Gene Functions	Ecological Relevance in Lowlands	Ecological Relevance in Uplands
Respiration and hypoxia tolerance	Globins, HIF pathway, lung morphogenesis regulators	Enhancing oxygen uptake and hypoxia signaling during prolonged submergence in hypoxic floodplains	Optimizing aerial breathing frequency and oxygen uptake efficiency in fast, oxygen-rich streams
Osmoregulation and ion transport	Aquaporins, Na+/K+-ATPases, chloride/bicarbonate exchangers	Coping with fluctuating conductivity and ionic imbalance	Stabilizing ion transport in clear, low-conductivity headwaters
Reproduction and developmental timing	Gonadotropin regulation, reproductive hormone signaling	Synchronizing spawning with flood pulses and rapid development in seasonal waters	Delayed reproduction and prolonged development adapted to seasonal rain patterns
Immune response and parasite resistance	Cytokine response, MHC-related genes, toll-like receptors	Defending against high pathogen loads in stagnant waters	Localized immune responses to specific microhabitat pathogens
Electric organ discharge modulation	Na+ leak channels, potassium modulators, ion channel tuning	Maintaining signal clarity in turbid, high-conductivity waters with high background noise	Signal modulation for species recognition and mate choice in clearwater, cluttered environments
Predator avoidance and stress response	Corticosteroid response, startle reflex circuits, CRH pathway	Avoiding predators in open, structure-poor habitats	Social coordination and vigilance in structurally complex habitats
General metabolic rate regulation	ATP synthesis, mitochondrial regulation, PGC-1Œ± pathway	Managing energetic stress in warm, low-oxygen conditions	Stable metabolism in cooler, oxygen-rich environments
Neurotransmission and sensory processing	Voltage-gated ion channels, GABAergic/glutamatergic balance	Enhancing electrolocation and short-range communication in low-visibility habitats	Maintaining precise electric signaling for social grouping and navigation

Circadian rhythm regulation	Melatonin signaling, circadian clock genes (e.g., Per, Cry)	Aligning behaviors with flood cycles and daylight shifts	Seasonal entrainment to rainfall and photoperiod
Muscle function and swimming performance	Myosin/actin regulators, calcium handling genes	Swimming efficiency in slow, silty, often deoxygenated waters	High-endurance swimming in fast-flowing or fragmented habitats

Lowland Speciation and Divergence Scenarios. Lowland lineages exhibited shallower divergence but strong ecological structuring. These patterns were first resolved through PCA of 3,090 filtered SNPs (Figure 4), which recovered three discrete genomic clusters**:** *E. multivalvulus* (Lineage 7, black), *E. varii* (Lineage 8, yellow)**,** and Lineage 9 (Negro/Branco clade, purple). STRUCTURE admixture analysis (K = 3) further supported this pattern (Figure 5). Upper Branco (Bomfim) individuals showed 100% assignment to Lineage 9, whereas individuals from the middle Branco and lower Negro showed 10–30% admixture with *E. varii*. In contrast, *E. multivalvulus* formed a distinct, unadmixed cluster. This asymmetric genetic pattern, combined with mito-nuclear discordance (e.g., *E. varii* mtDNA in Lineage 9 individuals), supports secondary contact with asymmetric gene flow rather than historical mitochondrial capture. RDA further revealed strong ecological differentiation (Figure 6). RDA1 explained a significant portion of constrained variance (Permutation ANOVA: $F = 11.02$, $p = 0.001$), separating whitewater (*E. multivalvulus*), blackwater (Lineage 9), and floodplain (*E. varii*) populations. Outlier SNPs were strongly associated with water type (Figures 7 & 8), consistent with divergent selection (Table 3). Interestingly, maximum likelihood phylogenies (IQ-TREE) failed to resolve Lineages 7 and 9 as distinct from *E. varii*, with these taxa appearing interspersed. This pattern likely reflects incomplete lineage sorting or recent divergence with gene flow, both of which are better captured by multivariate genomic approaches (PCA, STRUCTURE, RDA).

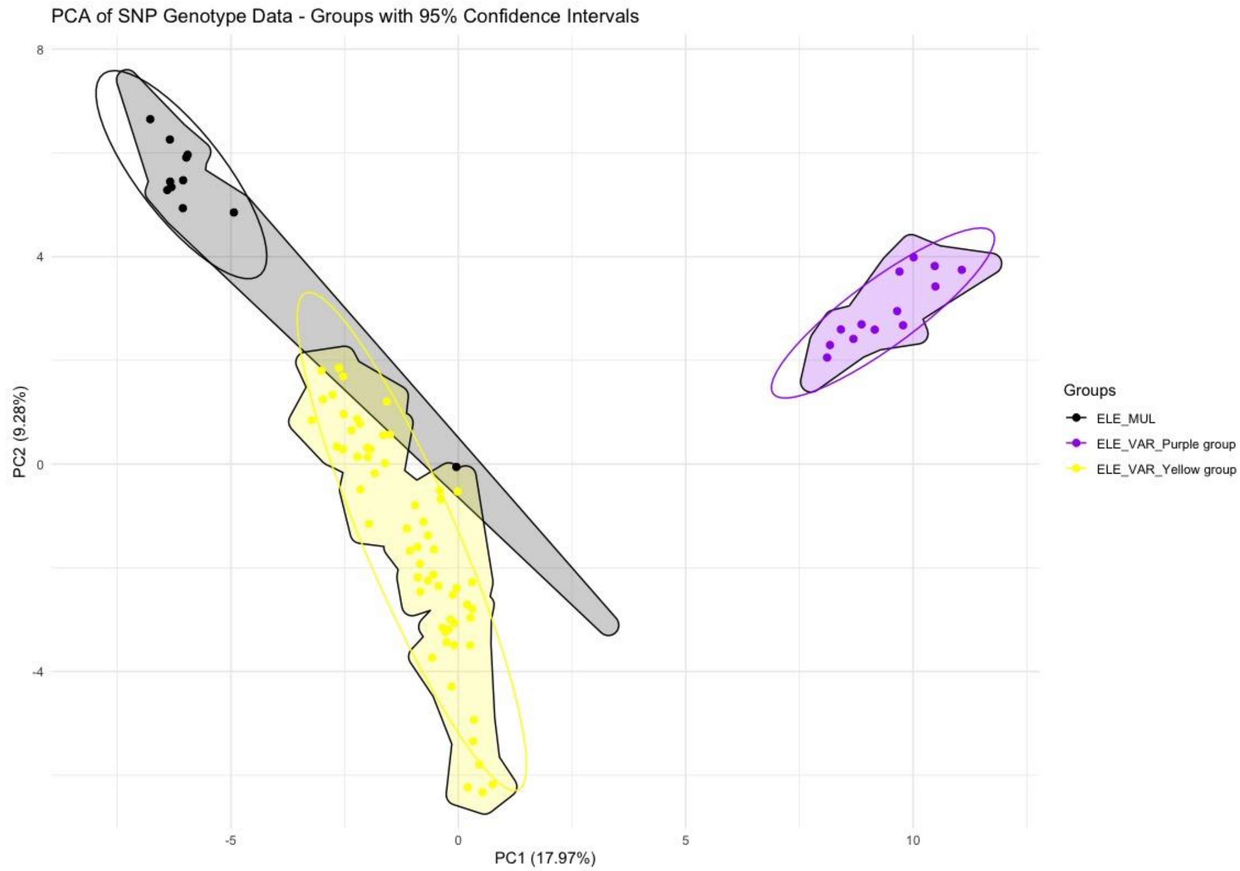


Figure 4. Principal Component Analysis (PCA) of lowland *Electrophorus* lineages. PCA based on 3,090 genome-wide SNPs reveals three distinct genetic clusters corresponding to: *Electrophorus multivalvulus* (Lineage 7, **black**) from whitewater headwaters (Ucayali and Pacaya Rivers), *Electrophorus varii* (Lineage 8, **yellow**), widespread in Amazonian whitewater floodplains, and Lineage 9 (**purple**), a cryptic blackwater-restricted lineage from the Negro and Branco Rivers. The non-overlapping clusters support strong genomic separation consistent with ecological divergence. The separation of Lineage 9, despite its shallow mtDNA divergence, indicates its independent evolutionary trajectory. In *E. multivalvulus*, the divergent individual plotted in *E. varii* corresponds to the Pacaya River (third from left to right in the STRUCTURE admixture bar plot), which has 65-70% ancestry from *E. varii*. In mtDNA, it evolved in *E. multivalvulus* clade. It suggests a recent introgression, that is, the exchange of genetic material between *E. multivalvulus* and *E. varii* that has occurred by secondary contact.

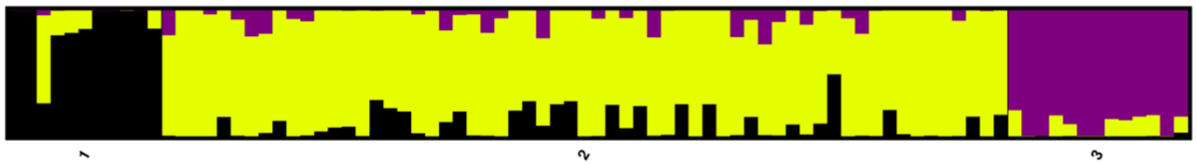


Figure 5. STRUCTURE admixture analysis (K = 3) of lowland *Electrophorus* lineages**.** STRUCTURE bar plot showing individual ancestry proportions across three genetic clusters: *Electrophorus multivalvulus* (Pop 1, **black**), *Electrophorus varii* (Pop 2, **yellow**), Lineage 9 (Pop 3, **purple**). Upper Branco (Bomfim) individuals exhibit 100% assignment to Lineage 9, while individuals from the middle Branco and lower Negro rivers show 10–30% admixture with *E. varii*. This asymmetric ancestry and mito-nuclear discordance support secondary contact with asymmetric gene flow between *E. varii* and Lineage 9. The clear clustering of *E. multivalvulus* further supports its status as an independently evolving lineage. Within *E. multivalvulus*, the individual from the Pacaya River (third from left to right in the STRUCTURE admixture bar plot) has 65-70% ancestry from *E. varii*. In mtDNA, it evolved in *E. multivalvulus* clade. It suggests a recent introgression, that is, the exchange of genetic material between *E. multivalvulus* and *E. varii* that has occurred by secondary contact.

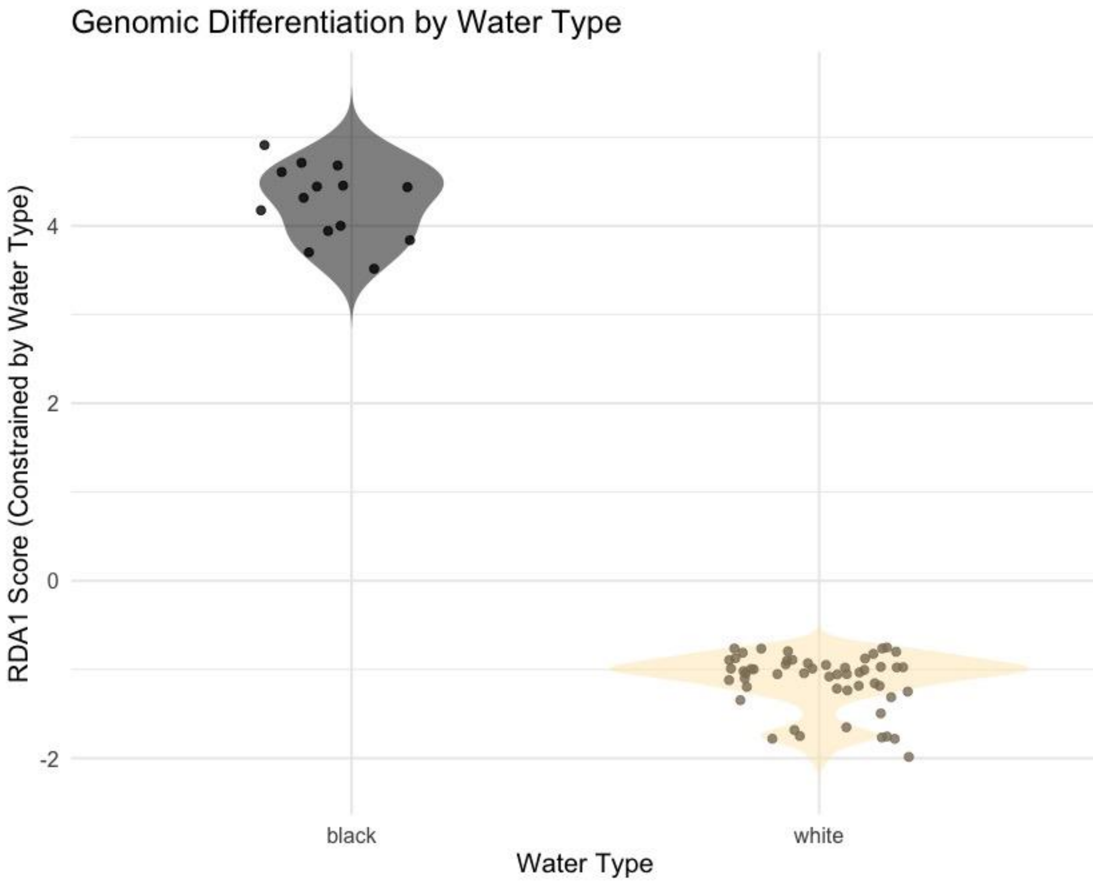


Figure 6. Genomic differentiation between blackwater and whitewater electric eel lineages based on RDA1 scores. Violin plot of individual RDA1 scores from redundancy analysis (RDA), constrained by water type. Each dot represents a sample. Blackwater lineage (Lineage 9) cluster at high RDA1 values, while whitewater lineages (Lineages 7 and 8) cluster around low to negative RDA1 scores. The clear separation of distributions indicates strong habitat-associated genomic differentiation (Permutation ANOVA: F = 11.02, p = 0.001), supporting ecological divergence between blackwater and whitewater taxa.

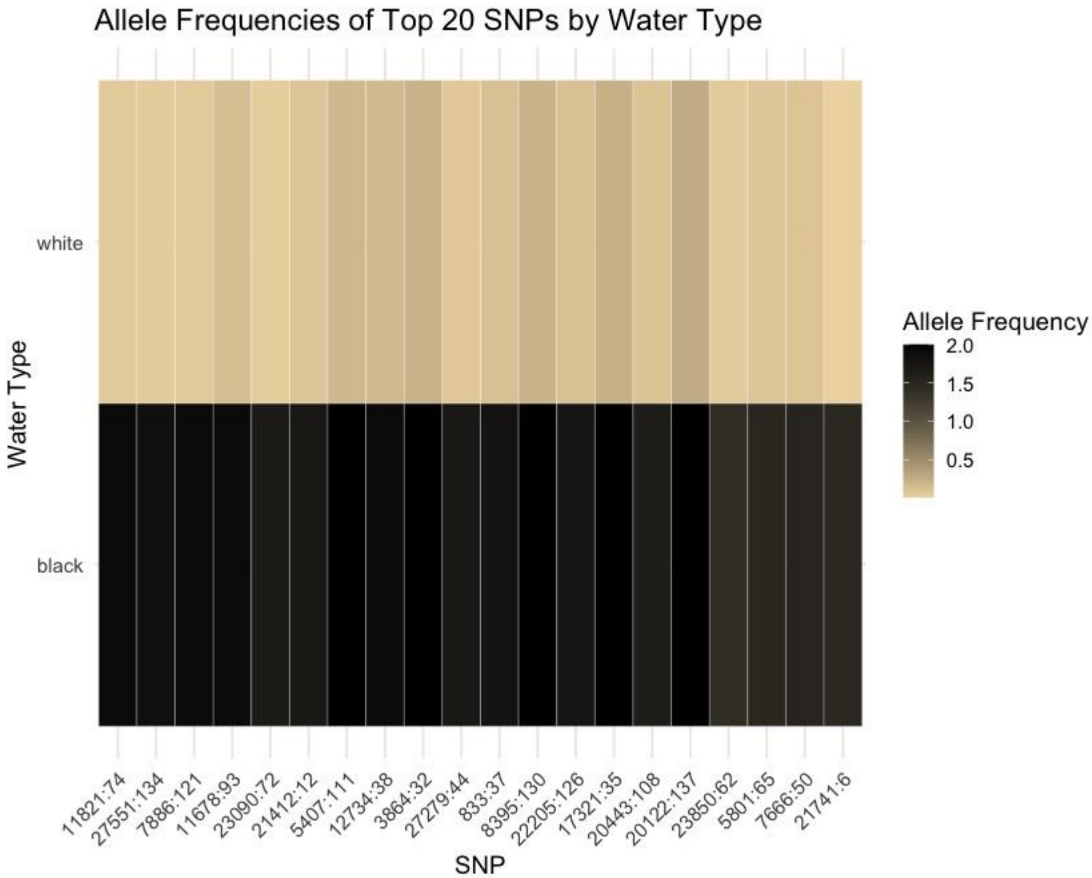


Figure 7. Heatmap showing the average allele frequencies for the 20 SNPs most strongly associated with habitat type based on loadings on RDA1. Rows indicate water types (whitewater vs. blackwater), and columns correspond to individual SNP loci. The color gradient represents allele frequency, from low (light cream) to high (black). Blackwater populations consistently exhibit higher allele frequencies, approaching fixation (allele frequency ≈ 2), at several outlier loci, indicating stronger habitat-associated selection. This pattern is likely driven by the extreme physicochemical conditions of blackwater environments, which are typically acidic (pH < 5), low conductivity (ion-poor), and rich in dissolved humic substances. These conditions impose unique physiological challenges that may favor alleles associated with osmoregulation, ion transport, detoxification, and sensory system adaptation. In contrast, whitewater habitats are more chemically buffered and variable, resulting in comparatively weaker and more heterogeneous selective regimes. This divergence highlights the role of ecological speciation across habitat gradients in shaping genomic differentiation within the electric eel complex.

Figure 8. The barplot shows the SNPs most strongly aligned with the first constrained axis (RDA1) from a Redundancy Analysis testing the effect of water type (black *versus* white) on SNP variation. All loci shown here have positive RDA1 loadings >0.36, indicating stronger allele frequency enrichment in blackwater populations. These SNPs are considered candidates under divergent selection due to environmental conditions specific to blackwater rivers, such as low pH, low ionic concentration, and high dissolved organic matter. When mapped to a reference genome, these SNPs can be annotated to identify gene regions potentially under selection. Candidate genes may include those involved in osmoregulation (e.g., ion transporters), acid-base balance, detoxification (e.g., cytochrome P450 enzymes), and sensory adaptation (e.g., olfactory and electrosensory receptors), which are critical for survival in chemically extreme blackwater environments.

Table 3. Hypothetical gene functions likely to be under selection in blackwater environments, based on known physiological challenges and typical ecological adaptations of aquatic organisms in low-pH, low-ion, and high-DOM waters

Gene Category	Gene/Protein Examples	Expected Function in Blackwater Adaptation
Osmoregulation	*NKA (Na^+/K^+-ATPase)*, *NKCC*, *Aquaporins*	Regulate ion uptake and water balance in ion-poor environments
Acid-base Regulation	*CA (Carbonic Anhydrase)*, *SLC4/26 family*	Maintain pH homeostasis under acidic conditions
Detoxification & Oxidative Stress	*CYP450s*, *GSTs*, *SOD*, *GPx*	Detoxify organic compounds and mitigate oxidative stress from humic-rich waters
Immune Response	*TLRs (Toll-like Receptors)*, *ILs*, *MHC*	Enhanced immune function due to pathogen-rich, organic-dense environments
Sensory Perception	*Olfactory receptors*, *Opsins*, *NaV*, *KCNQ*	Adaptation in electrocommunication or olfaction due to low visibility and high DOM
Electrosensory Genes	*scn4aa*, *cacna1a*, *kcna7a*	Voltage-gated ion channels for electric discharge and signal generation, potentially modulated by water chemistry
Energy Metabolism	*ATP synthase*, *NADH dehydrogenase*	Support energy-intensive osmoregulation and electrocommunication processes in harsh environments
Mucosal/Barrier Function	*Mucin genes (MUC)*, *claudins*	Maintain protective skin barrier and mucus layer in acidic, microbially rich waters

***** Actual gene identities must be confirmed by aligning our top SNPs (e.g., 11821:74, 27551:134, etc.) to a reference genome.

Divergence Scenarios for Lineages 7 and 8

- **Scenario 1: Divergence with Morphological Differentiation**
  If skeletal differences (caudal skeleton; Figure 9) are confirmed, divergence likely occurred in allopatry, during Miocene drainage fragmentation (Lake Pebas), followed by limited secondary contact and localized hybridization.

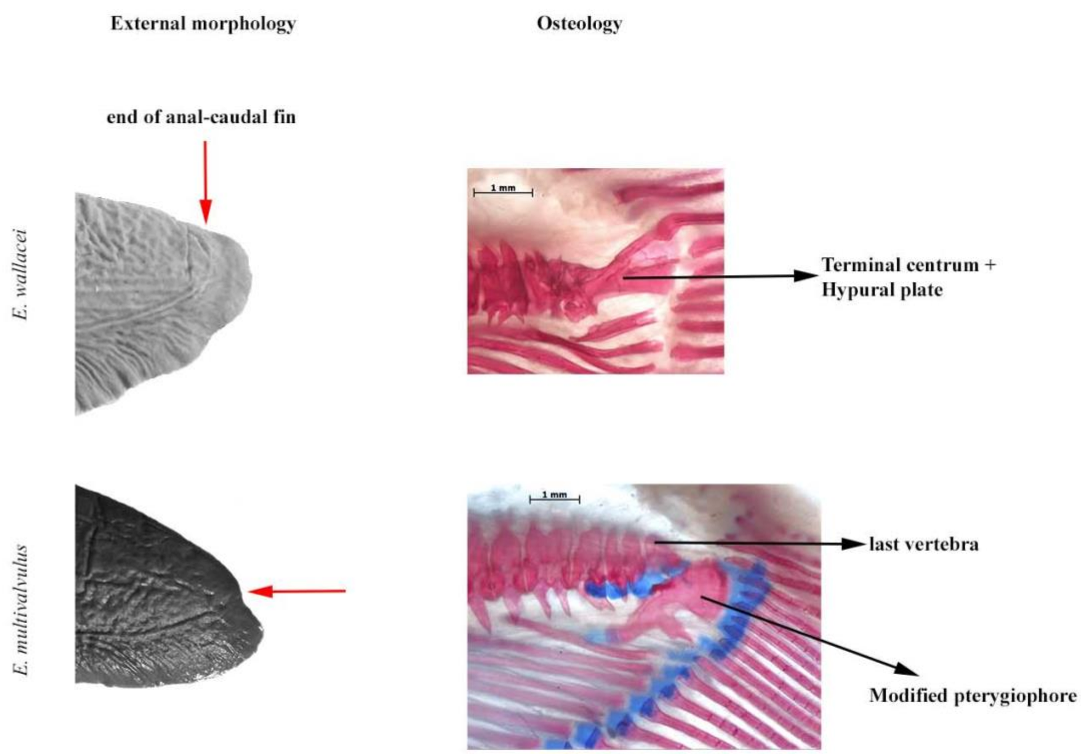


Figure 9. Caudal fin and skeleton in *E. varii* top and *E. multivalvulus* bottom. Note the absence of hypural plate in *E. multivalvulus*, and the presence of a modified pterogyophore.

- **Scenario 2: Cryptic Ecological Speciation**
  If morphological traits are not differentiated, divergence likely occurred via parapatric ecological speciation, with genomic islands maintained by selection in the face of gene flow.

Our results favor divergence during Miocene isolation within the Pebas system and secondary contact during eastward drainage integration. Morphological differentiation in the caudal skeleton and strong SNP-based structure support *E. multivalvulus* as an independently evolving lineage, now in contact with *E. varii* in the Pacaya River. This is a classic case of speciation with gene flow, driven by ecological divergence in a floodplain mosaic.

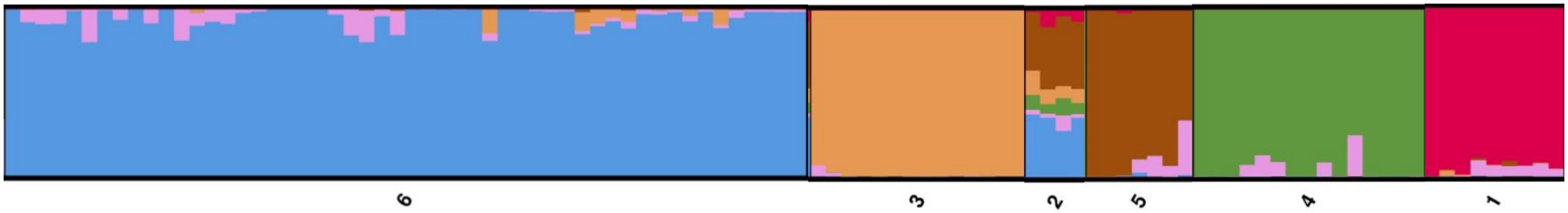
Upland speciation scenario. The six upland lineages (Pop 1–6) exhibit strong evidence of long-term allopatric divergence consistent with classical vicariant speciation across geologically stable regions of the Guiana and Brazilian shields. These lineages occupy discrete headwater drainages separated by major watershed boundaries (i.e., Madeira, Tocantins, Maroni), which act as persistent biogeographic barriers. Phylogenomic analyses (SNAPP, IQ-TREE) based on >110,000 SNPs reveal deeply structured relationships with reciprocal monophyly among these lineages (Figure 10), with Fst values ranging from 0.30 to 0.67 (Table 5) and mitochondrial COI divergences from 2.4% to over 7.6% (Tables 1 & 5). Moreover, PCA separates all six lineages, reinforcing their genetic independence (Figure 11). These genetic patterns are mirrored by ecological and behavioral divergence: upland lineages are highly philopatric, social as adults, breed during the rainy season, and often return to the same sites to forage. Despite this, STRUCTURE admixture analyses at K = 5 resolve five distinct genetic clusters, with partial admixture in the widespread Lineage 6 (*E. voltai*) and reduced resolution for Lineage 2, likely due to low sample size (N = 4). The tendency to underperform with the inclusion of low sample sizes in STRUCTURE admixture analyses is widely documented in the literature. For electric eels, this bias may be further emphasized by the unique physical trait of Lineage 2, which sets it apart from other lineages: it has an unelaborated lower lip, unlike the extremely bifurcated lower lips found in the other lineages. STRUCTURE admixture analyses at K = 6 is showed in the Figure 10.

Phylogenetic analyses based on SNP data indicate an initial divergence between *E. electricus* (found in the Essequibo, Cuyuni, Corantjin, Nickeri, Coppename, Suriname, and

Saramacca Rivers) and lineage 4 (upper Maroni), both of which are located in the Guiana Shield. Lineage 5 (upper Tocantins) diverged next, likely due to historical isolation within the central Brazilian Shield. This was followed by the diversification of lineage 2 (Orinoco), lineage 3 (upper Madeira), and *E. voltai* (which is widespread in both the Brazilian and Guiana shields), reflecting a north-to-south distribution across upland tributaries. These trends correspond with Miocene tectonic events (approximately 15–5 million years ago), including the Pebas system's formation and the Purus Arch's emergence, which created significant barriers to gene flow and facilitated allopatric divergence. The separation of Lineage 5 may indicate the early independence of Brazilian Shield drainages before the integration of the Amazon basin. Subsequent diversification among Lineages 2, 3, and 5 (*E. voltai*) was shaped by Andean uplift, the Vaupés Arch, and river capture events that disrupted headwaters. Fluctuations in sea levels during the Pliocene and Pleistocene (around 5 million years ago to the present) further altered hydrological conditions and led to periods of isolation, particularly impacting *E. voltai*. While upland habitats were relatively buffered from direct marine incursions, these climatic oscillations likely enhanced isolation-by-environment and reinforced the strong philopatry observed across these lineages.

This coordinated ecological divergence and long-term isolation suggest that the upland species arose via classic allopatric speciation, followed by environmental fine-tuning within their respective drainages.

A



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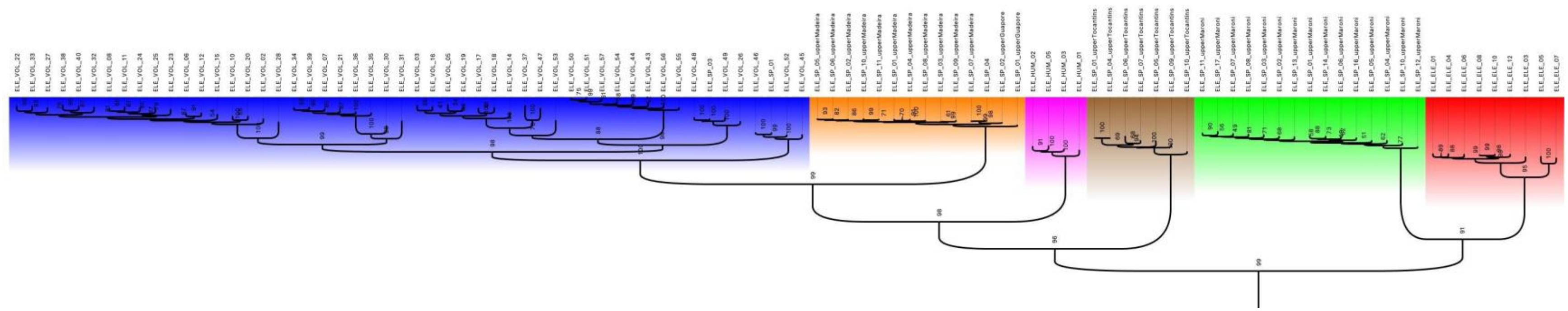

Figure 10. A. STRUCTURE admixture analysis for six populations, but it has chosen five lineages. However, Lineage 2 had four individuals, and several studies show that the STRUCTURE underperforms with samples per population. Lineage 1= *E. electricus*; Lineage 2 = Orinoco; Lineage 3 = upper Madeira, Lineage 4 = upper Maroni; Lineage 5 = upper Tocantins; Lineage 6= *E. voltai*. **B.** Consensus Maximum Likelihood of 110,793 SNPs with 1,000 ultrafast bootstraps in IQ-TREE.

Table 5. Summary of key genetic, ecological, and morphological metrics for the six upland lineages of *Electrophorus*, based on ddRAD-derived SNPs, COI mtDNA, and morphological assessments. Lineage IDs correspond to those used throughout the report. Colors match group assignments used in STRUCTURE and PCA figures. Metrics include the number of private SNPs, nucleotide diversity (π), Fst and K2P distances to *E. electricus* (Lineage 1), and qualitative clustering patterns in PCA, maximum likelihood phylogenies (IQ-TREE), and STRUCTURE. Additional notes highlight lineage-specific features such as hybridization, cryptic structure, or monophyly. Lower lip morphology is scored as bifid or unelaborated. All lineages are supported as valid species based on integrative evidence.

L	Drainage	Private SNPs	Pi(π)	Fst to Pop 1	mtDNA K2P vs Pop 1	PCA Cluster	IQ-TREE ML Placement	STRUCTURE Clustering	Notes	Lower Lip Morphology	Species Status
1	Guianas (Cuyuni to upper Saramacca)	4865	0.066			Distinct	Monophyletic, sister to Pop 5	Distinct cluster	Strongly supported across datasets	Bifid	Valid species
2	Orinoco	3283	0.064	0.576	0.0673	Partial overlap with Pop 5	Monophyletic, independent of Pop 5	Mixed with Pop 5 in major mode; separate in minor mode	Can be separated by morphology	Unelaborated	Valid species
3	Upper Madeira	2837	0.038	0.682	0.0682	Distinct	Monophyletic, sister to Pop 6	Distinct cluster	Deep divergence and clear SNP/MtDNA signal	Bifid	Valid species
4	Upper Maroni	7946	0.069	0.442	0.0763	Distinct	Monophyletic, basal to P3+P6	Distinct cluster	Strongly supported by SNP and mtDNA data	Bifid	Valid species
5	Upper Tocantins	6671	0.098	0.535	0.0695	Distinct	Monophyletic, sister to Pop 1	Distinct cluster	Diverged mtDNA + strong nuclear support	Bifid	Valid species
6	Brazilian + Guiana Shields (Tapajos, Xingu, Jatapu, etc.)	15077	0.118	0.386	0.066	Broad but structured	Monophyletic, sister to Pop 3	Single cluster in major mode; splits in minor mode (internal structure)	Potential cryptic species or ESUs within	Bifid	Valid species with structure

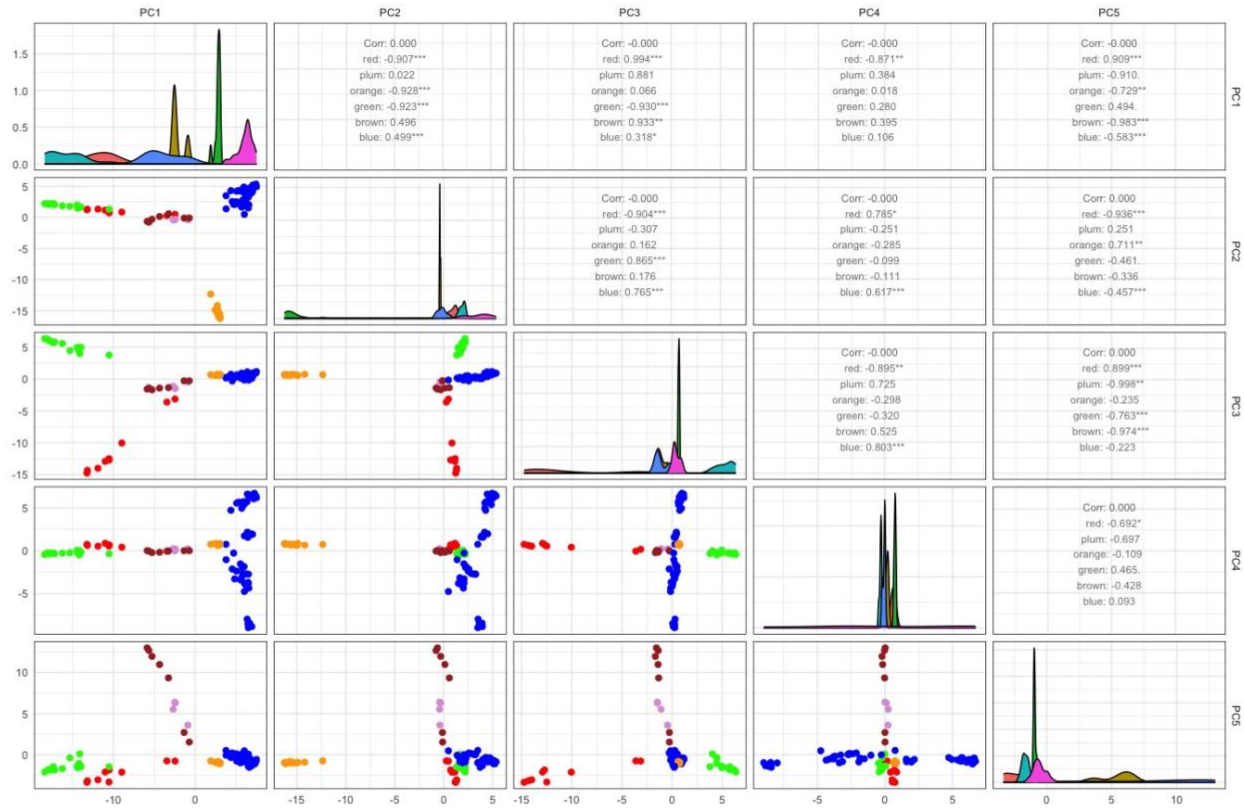

Figure 11. Pairwise PCA scatterplots (PC1–PC5) and density distributions for genomic variation across six lineages of Electrophorus. The diagonal panels display kernel density plots for each principal component (PC), while the lower triangle shows scatterplots of all pairwise combinations of PCs 1 to 5. The upper triangle presents Pearson correlation coefficients between each principal component and color-coded lineages. Asterisks denote statistical significance of correlations ($p < 0.05$ = *, $p < 0.01$ = **, $p < 0.001$ = ***). This multivariate analysis reveals strong genetic structure among the six lineages, with PC1 and PC2 capturing the most differentiation. Lineages show clear clustering patterns in multivariate space, indicating substantial genomic divergence consistent with species-level differentiation. Color legend for lineages: **Blue** – Linaege 6, *Electrophorus voltai*, from the Xingu–Tapajós, Denemi, Jatapu, upper Madeira, Javari, and Araguari rivers; **Brown** – Lineage 5 from the upper Tocantins River; **Orange** – Lineage 3 from the upper Madeira River; **Red** – Lineage 1, *Electrophorus electricus*, from the Essequibo, Cuyuni, Corantijne, Nickerie, Coppename, Suriname, and Saramacca rivers; **Pink** – Lineage 2 from the Orinoco River; **Green** – Lineage 4 from the upper Maroni).

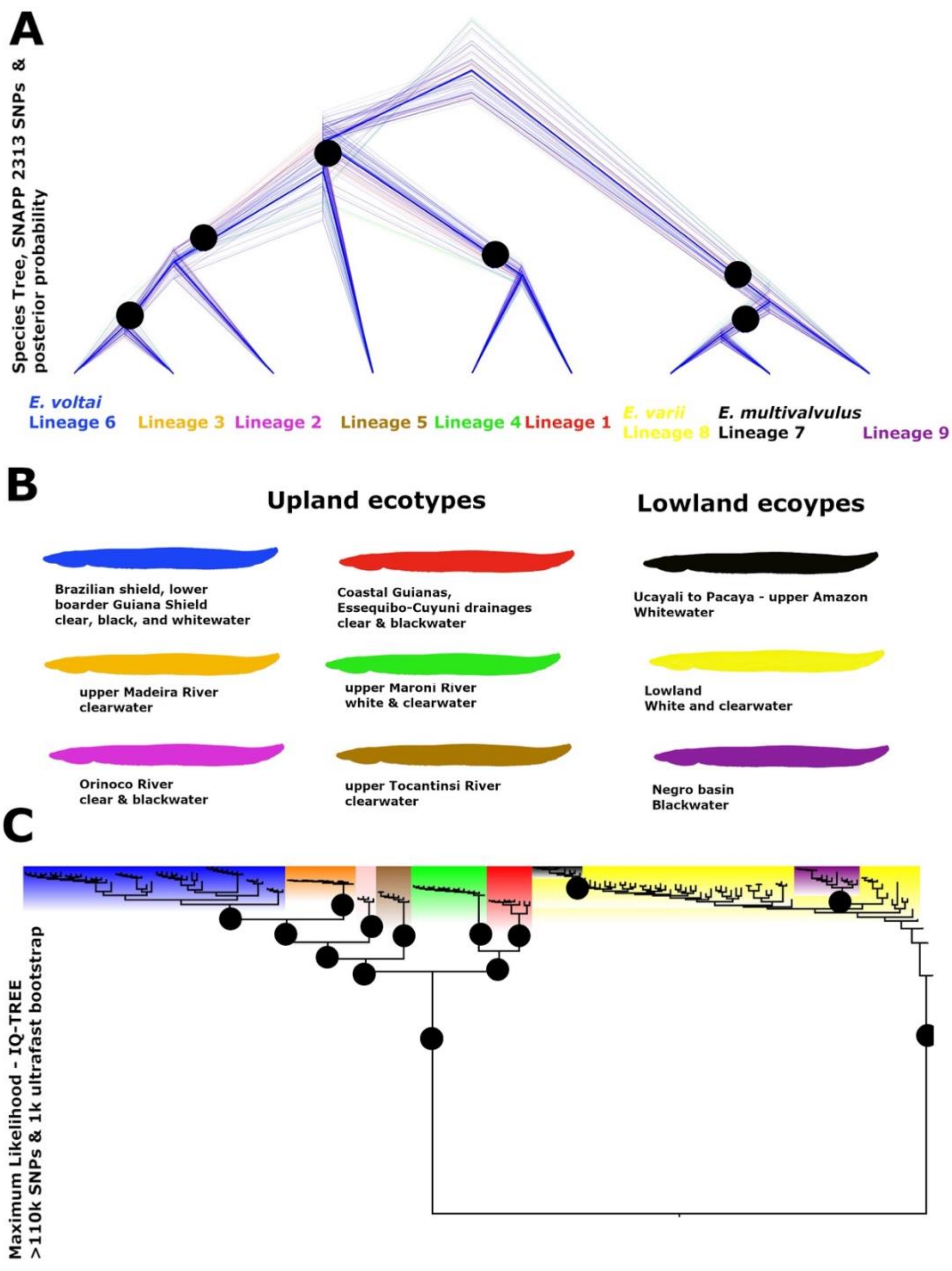

Figure 12. A. Species Tree of *Electrophorus* based on SNPs with SNAPP in Beast 25 samples and 2313 SNPs; **B.** Lineage distribution and water type; **C.** Maximum likelihood with 1,000 ultrafast bootstraps of 110,783 SNPs and 175 samples performed in IQ-TREE. Tree rooted *a posteriori* at midpoint.

Phenotypic conservatism

Anatomical conservatism in electric eels reflects the profound functional constraints imposed by their unparalleled electrogenic capabilities. Unlike any other gymnotiforms, electric eels possess three specialized electric organs—the Main, Hunter's, and Sachs'—that occupy nearly 80% of their elongated body and are finely tuned for both low-voltage electrolocation and high-voltage predatory and defensive discharges, reaching up to 860 volts in *E. voltai*. Their streamlined, cylindrical form is not merely ancestral but essential: the longitudinal arrangement of electrocytes enables efficient spatial summation of electric potentials, a process susceptible to body geometry. For example, during predatory strikes, eels adopt a *U-shaped* posture, aligning head and tail around the prey, doubling the voltage delivered. During coordinated group hunting, they employ *S-shaped* movements to synchronize powerful electrical discharges and collectively overpower their prey. These electromechanical behaviors demand exceptional neuromuscular control and structural precision, leaving little room for morphological deviation. Despite occupying a broad ecological range within the Greater Amazonia, species of *Electrophorus* maintain a remarkably conserved body plan. This suggests that their ability to thrive across diverse habitats depends not on morphological innovation, but on the plasticity of electric behavior and sensory tuning. Stabilizing selection preserves this electrogenically optimized anatomy across all lineages, resulting in striking morphological uniformity despite deep genetic and ecological divergence. Electric eels thus exemplify how strong performance-based constraints can drive anatomical stasis even in the face of widespread ecological adaptation.

Preliminary Conclusions

Hypothesis 1: Do major mtDNA lineages correspond to independently evolving species?
Our phylogenomic dataset confirms that most major mtDNA clades represent independently evolving lineages with consistent genomic clustering in PCA and STRUCTURE, reciprocal monophyly in SNAPP species trees, and elevated Fst values. Notably, upland lineages (e.g., *E. electricus*, *E. voltai*) show deep mtDNA and nuclear divergence, consistent with species-level separation. However, in lowland lineages (*E. varii*, *E. multivalvulus*, Lineage 9), mtDNA alone failed to detect divergence, particularly for Lineage 9, highlighting the limitations of single-locus approaches. Multivariate SNP-based analyses resolved this cryptic lineage, reinforcing the need for genome-wide data in recent radiations.

Hypothesis 2: Does ecological divergence between upland and lowland habitats drive speciation?

We recovered a deep basal split between upland and lowland ecotypes, each associated with distinct ecological regimes—clearwater headwaters vs. floodplain white/blackwaters—and divergent life-history traits (breeding season, sociality, site fidelity). RDA revealed that habitat type explains a significant portion of genomic variation (RDA1 $F = 162.15$, $p = 0.001$), and SNP outliers are enriched for habitat-specific divergence. These results support ecological speciation along the upland–lowland axis, with environmental pressures maintaining reproductive isolation despite partial geographic overlap in some drainages.

Hypothesis 3: Does morphological stasis mask cryptic diversification?
Despite marked genetic and ecological divergence among the nine *Electrophorus* lineages, all share a conserved external morphology shaped by the functional demands of electrogenesis. The anatomical uniformity, especially the elongated body and electro-organ structure, likely reflects strong stabilizing selection. Nonetheless, multiple cryptic species were uncovered, particularly in the lowlands, where PCA and STRUCTURE revealed genetically and ecologically distinct lineages indistinguishable by gross morphology. In the case of *E. multivalvulus*, caudal skeleton distinguished it from *E. varii*.

Our analyses of electric eels reveal that *Electrophorus* is a radiation of ecologically and genetically distinct lineages evolving under strong functional constraints. A deep basal split between upland and lowland ecotypes aligns with divergent environmental regimes, social behavior, breeding season, and genomic architecture, supporting ecological speciation despite overlapping geography. We show that while mtDNA captures deep divergence in upland taxa, only genome-wide SNPs resolve cryptic diversity in lowland floodplains. Strikingly, all nine lineages maintain a conserved external morphology, reflecting the stabilizing selection imposed by their extreme electrogenic anatomy. Together, these findings uncover four modes of speciation in *Electrophorus*: 1, ecological between upland and lowland ecotypes; 2, allopatric in the upland lineages; 3. ecological with gene flow across whitewater–blackwater gradients, and 4. cryptic speciation between *E. multivalvulus* and *E. varii*, with limited introgression. All unfolding beneath the surface of morphological stasis. These preliminary results highlight how ecological divergence, rather than geography or phenotype alone, can drive and maintain species boundaries in one of the most iconic vertebrates.

Next steps:

1. Phylogenetic analyses with the immediate outgroup, *Gymnotus*;
2. Whole-Genome and SNPs outliers alignment (Functional annotation of outlier SNPs detected in RDA will identify genes under divergent selection across water types, e.g., those involved in ion transport).